

GESER DUGAROV, PhD | Software Engineer | Distributed Data Systems & AI Data Infrastructure

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SUMMARY

Software engineer building petabyte-scale lakehouse and AI data infrastructure at Huawei. Designed and implemented a two-tier distributed cache for Lance; currently developing a query-adaptive ANN index targeting tens of billions of vectors. As part of this role, authored 50 merged PRs to [Apache Hudi](#) and three to [Lance](#). Research background informs a rigorous approach to system design, benchmarking, and performance validation. Creator of open-source agentic software-development tooling for reproducible issue-to-PR and develop-review workflows with human-in-the-loop (HITL) handoff, monitoring, and cost telemetry.

TECHNICAL SKILLS

Java, Python, Rust (some experience), PostgreSQL, Apache Hudi, Lance, Apache Spark, Apache Flink, Ray, vector search and ANN indexing, distributed systems, performance engineering, LLM agent orchestration, multi-agent workflows

WORK EXPERIENCE

May 2023 – Present **Software Engineer / Big Data Engineer,**
[Huawei](#)

Develop core lakehouse and AI data infrastructure for Huawei Cloud.

- Designed and implemented a distributed cache for Lance, pairing an in-memory L1 with a persistent L2 on per-worker NVMe and prewarming IVF partitions to avoid repeated object-storage reads.
- Building a distributed, incrementally updatable vector index targeting tens of billions of vectors using query-adaptive ANN search.
- Simplified Hudi setup by automating recommended defaults for common scenarios in a system with 900+ configuration parameters.
- Implemented partition-level TTL, automating data retention and management of cloud storage costs.

Selected upstream open-source contributions (delivered as part of the Huawei role)

[Apache Hudi](#) | Jan 2024 – Present | [50 merged PRs](#)

- Optimized serialization of Flink stream records in Hudi, cutting end-to-end write time by up to 26% and reducing data transferred between operators by 29–36% in TPC-H benchmarks ([design doc](#), [main changes](#)). Released in [Hudi 1.0.2](#).
- Implemented four targeted optimizations ([#12054](#), [#12104](#), [#12113](#), [#12120](#)), improving Flink stream-write throughput by 10% and reducing garbage-collection overhead by 30% in internal benchmarks. Released in [Hudi 1.0.1](#).
- Leading RFC-98 for Spark DataSource V2 read support; the design and Copy-on-Write implementation are under review.

[Lance](#) | May 2026 – Present | [3 merged PRs](#)

- Authored three merged Rust PRs in Lance's vector-index build path, improving memory bounds and temporary-resource lifecycle in long-running index jobs.

Feb 2022 – May 2023 **Software Engineer / ML Engineer,**
Digital Research (computer vision startup)

- Designed an event-driven architecture and implemented the image-processing backend in Python for a truck-monitoring system handling ~20,000 images/day; in production, the system reduced fleet idle time by 12%.
- Built a customer-facing reporting and data-visualization UI and an internal monitoring UI.

PERSONAL PROJECTS

Apr 2026 – Present **Open-source: Agentic Development Tooling**

Self-directed work on AI-powered developer tooling: multi-agent implementation and review workflows with human-in-the-loop handoff, reproducible execution, monitoring, and cost tracking.

- Built [agent-orchestrator](#), a Python-based, GitHub issue-driven workflow runner that decomposes an issue, runs an implementer agent in an isolated git worktree, opens a PR, invokes a separate review agent, and notifies a human when the PR is ready to merge.
- Shipped eight agent-orchestrator releases from v0.1.0 (May 2026) to v0.8.0 (Jul 2026), adding workflow-state persistence in GitHub issues, operator-controlled pause/resume, PostgreSQL-backed usage and cost analytics, and an agent-trajectory viewer.
- Recorded 5,100+ agent runs across 12 repositories since May 2026, including 1,500+ review-agent runs; usage was equivalent to \$11K+ at API list prices.
- Built [agent-develop-review-loop](#), a Bash-based develop-review loop with HITL handoff, resumable runs, and cost telemetry parsed from agent usage metadata.

EDUCATION

PhD, Geophysics, Trofimuk Institute of Petroleum Geology and Geophysics SB RAS

MSc, Computational and Applied Mathematics, Novosibirsk State University

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